Advanced extensions

Optional dynamics

> **Optional dynamics — not examined.** This section supports later study but is not required for the pre-course, checkpoints, two-paper mock examination or two-paper entry examination.

This material sits outside the eight-week core and its 9–11 hour weekly workload. It is not a ninth week, is not a prerequisite for any required outcome and must not displace core study. You should however study this material at some point in the future.

First-order difference equations (`R1`)

The optional outcome is to solve a first-order linear difference equation and classify the stability of its steady state.

Reading: `MATH-R1-OSB-OPT` / `OSBORNE_MM` — Martin J. Osborne, *Mathematical Methods for Economic Theory*, Section 9.1, "First-order difference equations" (about 45 minutes).

Skip: Section 9.2 in full. Do not study second-order equations, systems, software material or any assessed application from this route.

Elementary first-order differential equations and phase lines (`R2`)

The optional outcome is to solve elementary separable and linear first-order differential equations and use a phase line to classify steady-state stability.

Reading: `MATH-R2-OSB-OPT` / `OSBORNE_MM` — Martin J. Osborne, *Mathematical Methods for Economic Theory*: Section 8.1, "Introduction", only the first-order existence/stability definitions in Section 8.2, Section 8.3 on separable equations, Section 8.4 on linear first-order equations, and Section 8.5 on phase diagrams for autonomous equations (about 95 minutes).

Skip: in Section 8.2, the existence proof; Sections 8.6–8.7 in full, including second-order equations and systems; all software; and every assessed use.

Advanced later extensions

If you become interested in economic theory and or dynamics, you may want to also study the following over the course of your PhD.

- **Real analysis** — a possible later foundation for rigorous graduate economic theory.
- **Dynamic programming** — a possible later method in graduate economic dynamics.

If this is the case, contact the PhD Coordinator for suggestions about books and material to study.